

Calculating the Usage Probabilities of Statistical Usage Models by Constraints Optimization

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ABSTRACT

The systematic generation of test cases from statistical usage models has been investigated recently for specific application domains, such as wireless communications or automotive applications. For Markov chain usage models, the expected usage of a hardware/software system is represented by transitions between usage states and a usage profile, meaning probability values that are attached to the state transitions.

In this paper, we explain how to calculate the profile probabilities for the Markov chain usage model from a set of linear usage constraints and by optimizing a convex polyhedron that represents the constrained solution space. Comparing the computed probability distributions of our polyhedron approach with the maximum entropy technique, which is the main technique used so far, illustrates that our results are more obvious to the intended constraint semantics. In order to demonstrate the applicability of our approach, workflow testing of a complex RIS/PACS system in the medical domain was carried through and has provided promising results.

Categories and Subject Descriptors

D.2.5 [Software Engineering]: Testing and Debugging—*usage profile, optimization*

General Terms

Optimizing Usage Probabilities

Keywords

Statistical usage testing, Markov chain usage model, profile generation, usage constraints, polyhedron optimization, medical case study.

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1. INTRODUCTION

Statistical software testing has experienced a large growth of interest recently [KO00, DZ03, Pro05, SP07, Dul08]. As distinct from other model-based testing activities [Ber07], the statistical testing approach benefits from usage-driven testing and the possibility to make statistical inferences about the system, based on a set of statistically sampled test cases (Fig. 1).

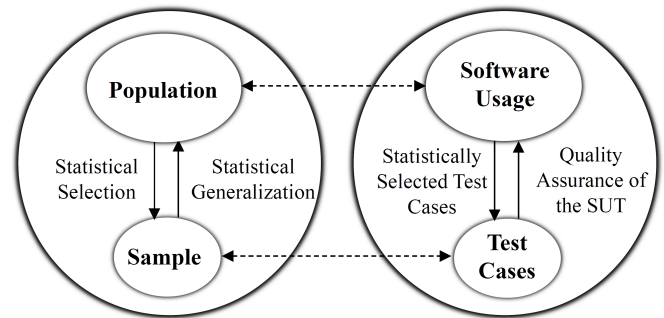


Figure 1: Statistical Testing versus Sampling.

Statistical testing is focussing on the systematic generation of test cases from a given usage model and to provide a basis for quality assurance in a model-driven testing process. In this context, a MCUM (Markov chain usage model [WP93, WPT95]) specifies the estimated usage of the SUT (Software/System Under Test) in its intended environment.

The MCUM is a directed graph that consists of states and transitions between states [WP93] and is used to define all possible usage scenarios that are derived by randomly traversing the arcs between two dedicated start and end states. Transitions possess probabilities that are collected in usage profiles in order to reflect usage choices that a user will have while interacting with the SUT (Fig. 2).

Providing more than one usage profile for a given Markov chain structure is also allowed. That way, the statistical selection of test cases reflects distinct properties of different roles or user classes, for instance experts or normal users, in interaction with the SUT.

Markov chains are mathematically characterized by tran-

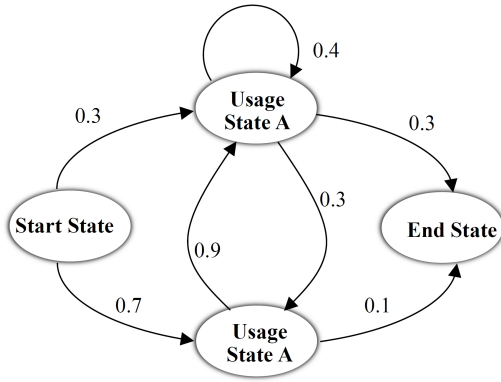


Figure 2: Example for a Markov chain usage model.

sition matrices¹. The i -th row of the j -th column of the transition matrix contains the transition probability for leaving usage state i and entering usage state j as depicted in Fig. 3.

	Start State	Usage State A	Usage State B	End State
Start State	0.0	0.3	0.7	0.0
Usage State A	0.0	0.4	0.3	0.3
Usage State B	0.0	0.9	0.0	0.1
End State	0.0	0.0	0.0	0.0

Figure 3: Transition matrix corresponding to the MCUM in Figure 2.

Each path of the MCUM represents a possible use of the SUT, and corresponding transition probabilities define the mean rate at which a customer will traverse this part of the SUT during the application.

Analytical computations can be applied even before test cases are executed in order to obtain characteristic test metrics. Examples are the Kullback-Leibler divergence (see section 5) for comparing usage profiles or the mean number of test cases to reach a certain confidence level [Pro00].

After performing sampled test cases in the test execution environment, test statistics are available and statistical inferences based on the results can be made. One of the most interesting facilities, statistical testing offers, is the estimation of the achieved reliability for the SUT [Say99].

Because all MCUM metrics heavily rely on the calculated probabilities for a given profile, methods for deriving correct transition probabilities are a critical part in creating a MCUM [WP00]. In this paper we focus on the computation of probability distributions from a given set of probability constraints by applying a polyhedron approach.

¹The transition matrix is often called stochastic matrix since the matrix elements are probability values.

2. DERIVATION OF USAGE PROFILES

2.1 Motivation

After all structural components, i.e. states and arcs of the MCUM, have been identified [BD05], appropriate transition probabilities have to be obtained. Ideally, for every state of the MCUM, the exact probability distribution of the outgoing arcs of each state is known and fixed. Unfortunately, this is often not the case. For example, in the medical domain very large and complex RIS/PACS systems have to be tested, where the usage of the system is subject to diverse influences [Hol08]. American Image Centers, for example, show a typical similar system usage and can be treated as one customer type. The behavior of this customer type is therefore mapped to a single MCUM.

2.2 General Strategies for Profile Derivation

In [WP93], Whittaker proposes three possible techniques to compute MCUM transition probabilities.

2.2.1 Uninformed Strategy

This is the most simple but also least suitable strategy for creating usage profiles. All outgoing arcs of a state within the usage profile are assigned equal probability values. While this approach does not presume any information about system usage, it can be applied if no such information is available.

2.2.2 Informed Strategy

Informed probability generation presumes collecting field data, which is then mapped to a usage profile. Hence, this method represents the most accurate way of gaining information about system usage. Field data may be derived from direct logging of user interaction use or screen recording, and can afterwards be evaluated or directly mapped to the model structure.

2.2.3 Intended Strategy

Probability values are defined by estimating the intended use of the system and strongly depend on the quality of the information sources. We preferred this strategy because neither changes on the system nor any additional technical resources are necessary. Development teams often possess a multitude of information sources for the required customer specification. In addition, application specialists, customer support and application trainers generally know the different customers and their specific way of working very well.

3. CONSTRAINT-BASED STRATEGY

Since it can not be assumed that customer experts do also have expert knowledge about statistics and MCUM theory, the process of information acquisition must be arranged in the most informal way as possible. This implies that customer experts are allowed to describe their knowledge in terms of *constraints* [WP00]. Instead of assigning fixed numbers to the MCUM transition probabilities, they have to postulate relationships between them. These constraints can further be automatically transformed to fixed probabilities by using algebraic techniques.

Referring to the simple state example shown in Fig. 4, a customer expert has several possibilities to formulate his knowledge.

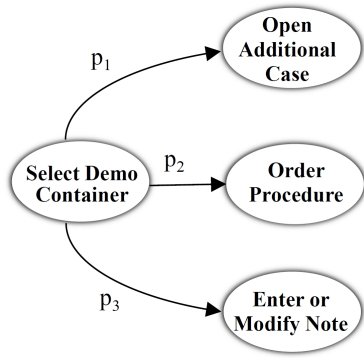


Figure 4: MCUM for constraint specifications.

1. Area Definition

"Having selected a demo container, the fraction of time a radiologist would order a procedure is definitely bigger than 20%."

This would result in $p_2 > 0.2$.

2. Interval Definition

"Having selected a demo container, the fraction of time a radiologist would open an additional case is in between 50% and 80%."

This would result in $p_1 \geq 0.5$ and $p_1 \leq 0.8$.

3. Exact Definition

"Having selected a demo container, the fraction of time a radiologist would enter or modify a note is exactly 25%."

This would result in $p_3 = 0.25$.

4. Dependent Definition

"Having selected a demo container, the fraction of time a radiologist would order a procedure is twice as big as opening an additional case."

This would result in $p_2 = 2 \cdot p_1$.

Constraints can be further classified as follows:

- **Structural constraints** are implicitly given by the model structure:
 - Probability values $p_{i,j}$ of all outgoing arcs $a_{i,j}$ of a state s_i must sum up to one, i.e. $\forall i : \sum_j p_{i,j} = 1$
 - Probability values $p_{i,j}$ must be in the interval $[0..1]$, i.e. $\forall i, j : 0 \leq p_{i,j} \leq 1$
- **Usage constraints** are given to define the intended system usage and cannot be handled automatically.

Constraints reduce the solution space when searching for a suitable probability distribution for the MCUM. This is demonstrated by re-using the previous constraints definitions referring to Fig. 4:

- Taking the constraints (iii) and (iv) ($p_3 = 0.25$, $p_2 = 2 \cdot p_1$) in addition to the structural constraints ($p_1 + p_2 + p_3 = 1$, $p_1 \geq 0$, $p_2 \geq 0$, $p_3 \geq 0$, $p_1 \leq 1$, $p_2 \leq 1$, $p_3 \leq 1$), the correct probability values are implicitly given by $p_1 = 0.25$, $p_2 = 0.5$, $p_3 = 0.25$.
- Considering constraints (i) and (ii) instead of (iii) and (iv), there exists an infinite set of possible solutions. Probability values $p_1 = 0.7$, $p_2 = 0.25$, $p_3 = 0.05$ are valid as well as $p_1 = 0.5$, $p_2 = 0.3$, $p_3 = 0.2$.

Obviously, objective functions have to be defined so that they deliver a single probability distribution in case that the number of possible solutions is greater than one.

3.1 Maximum Entropy Approach

In [PWW00], it was argued to use the entropy of the probability distribution as objective function and to maximize this entropy afterwards.

The state entropy $h(s_i)$ of a state s_i with n outgoing arcs $a_{i,j}$ and their corresponding probabilities $p_{i,j}$ with $j = 1..n$ is defined as:

$$h(s_i) = - \sum_{j=1}^n p_{i,j} \cdot \log_2(p_{i,j}) \quad (1)$$

This seems reasonable because the entropy is a measure of uncertainty and therefore guarantees that the information content is limited exclusively by the defined constraints. Considering the entropy function without defining constraints, the maximum is found when the probability values represent a uniform distribution, which also matches the concept of the uninformed strategy 2.2.1.

3.1.1 Constraints as Equalities

Let us consider a constrained, non-linear optimization problem with equality constraints first. Optimization problems of the form

$$\max f(p_1, \dots, p_k)$$

with constraints

$$\begin{aligned} g_1(p_1, \dots, p_k) &= 0 \\ &\vdots \\ g_m(p_1, \dots, p_k) &= 0 \end{aligned} \quad (2)$$

can be solved by using the Lagrange Method as proposed in [PWW00]. Constraints, as discussed in section 3, can easily be converted to this form by simple transformations. Having the operational function

$$f(p_1, \dots, p_k) = - \sum_{i=1}^k p_i \cdot \log_2(p_i), \quad (3)$$

the structural constraint

$$g_1(p_1, \dots, p_k) = -1 + \sum_{i=1}^k p_i, \quad (4)$$

and $m - 1$ usage constraints

$$g_j(p_1, \dots, p_k) = 0, \quad j = 2..m, \quad (5)$$

the Lagrange function Λ can be defined as follows

$$\Lambda(p_1, \dots, p_k, \lambda_1, \dots, \lambda_m) = f(p_1, \dots, p_k) + \sum_{j=1}^m \lambda_j \cdot g_j(p_1, \dots, p_k) \quad (6)$$

where λ_j are Lagrange Multipliers. The gradient of Λ is given by

$$\nabla_{p_i, \lambda_j}(\Lambda) = \frac{\partial \Lambda}{\partial (p_i, \lambda_j)} = \begin{pmatrix} \log_2 p_i & \sum_{t=1}^m \frac{\partial \lambda_t g_t}{\partial p_t} \\ g_j \end{pmatrix} \quad (7)$$

Assuming only linear constraints, the constraint functions can be written as

$$g_t(p_1, \dots, p_k) = \sum_{i=1}^k (a_{t,i} \cdot p_i) + c_i, \quad \text{with } a_{t,i} \text{ and } c_i \in \mathbb{R} \text{ and } t \in [1..m] \quad (8)$$

where $a_{t,i}$ and c_i are fixed constants, resulting from the previous constraint definitions. The gradient of Λ can be simplified to

$$\nabla_{p_i, \lambda_j}(\Lambda) = \frac{\partial \Lambda}{\partial (p_i, \lambda_j)} = \begin{pmatrix} \log_2 p_1 + \sum_{t=1}^m a_{t,1} \cdot \lambda_t \\ \vdots \\ \log_2 p_k + \sum_{t=1}^m a_{t,k} \cdot \lambda_t \\ \sum_{t=1}^k (a_{1,t} \cdot p_t) + c_1 \\ \vdots \\ \sum_{t=1}^k (a_{m,t} \cdot p_t) + c_m \end{pmatrix} \quad (9)$$

In order to obtain the interesting values of Λ , $\nabla(\Lambda)$ is set to zero. This leads to a system of $(k+m)$ linear equations with $(k+m)$ unknowns:

$$\begin{aligned} \log_2 p_1 + a_{1,1} \cdot \lambda_1 + \dots + a_{m,1} \cdot \lambda_m &= 0 \\ &\vdots \\ \log_2 p_k + a_{1,k} \cdot \lambda_1 + \dots + a_{m,k} \cdot \lambda_m &= 0 \\ a_{1,1} \cdot p_1 + \dots + a_{1,k} \cdot p_k &= 0 \\ &\vdots \\ a_{m,1} \cdot p_1 + \dots + a_{m,k} \cdot p_k &= 0 \end{aligned} \quad (10)$$

Solving this equation system results in a probability set $(p_1, \dots, p_k)$ satisfying the maximum entropy principle.

3.1.2 Constraints with additional inequalities

The method of Lagrange Multipliers can be further extended to handle additional inequality constraints. The Karush-Kuhn-Tucker Conditions (KKT [BG97]) represent a generalization of the method of Lagrange Multipliers. Let us consider the same optimization problem as before, extended with n additional inequalities:

$$\begin{aligned} h_1(p_1, \dots, p_k) &\leq 0 \\ &\vdots \\ h_n(p_1, \dots, p_k) &\leq 0 \end{aligned} \quad (11)$$

The KKT conditions state that p is a global minimum² of $f(p)$ if

$$\nabla f(p) + \sum_{i=1}^m \mu_i \nabla g_i(p) + \sum_{j=1}^n \nu_j \nabla h_j(p) = 0 \quad (12)$$

under the conditions

1. $g_i(p)$ are affine functions,
2. $h_j(p)$ are convex functions,
3. $\forall i = 1..m, j = 1..n : \nu_j \geq 0$ and μ_i exist,
4. $\forall j = 1..n : \nu_j \cdot h_j(p) = 0$.

Solving (12) can be done in analogy to solving (10).

4. POLYHEDRON APPROACH

4.1 Motivation

The maximum state entropy strategy faces a major drawback: maximizing the state entropy (1) will result in transition probabilities for outgoing arcs of each state that are always as uniform as possible. For example, referring to Fig. 4 and using the maximum state entropy (2), the single constraint $p_1 \geq 2 \cdot p_2$ would lead to the same result as the constraint $p_1 = 2 \cdot p_2$ would do.

However, the customer expert, who formulated the constraint, probably had in mind p_1 to be *mostly* bigger than twice the amount of p_2 and to handle the equality only as a marginal value. Thus, maximizing the state entropy may lead to inadequate solutions in certain cases. We therefore advocate the usage of a more appropriate method for calculating constrained probabilities, while assuring the most proper constraints semantics.

4.2 Computing the Probability Distribution

The key idea is as follows. Without any constraints, the solution space of the searched probability distribution for a state with k outgoing arcs would be $\mathbb{R}^k$. Of course, at least the structural constraints and, if stated, any usage constraints, restrict the solution space.

Equality constraints can be considered as restrictions to affine subspaces of lower dimensions within $\mathbb{R}^k$ (see Fig. 5) while inequalities result in restrictions to half spaces, as shown in Fig. 6. Regarding lots of constraints will lead to a convex polyhedron that is representing the constrained solution space. Based on this polyhedron, a suitable solution for the MCUM transition probabilities is obtained.

Again, let $g_k(p)$ be the set of equality constraints, and $h_l(p)$ the set of inequality constraints where p is the vector of all outgoing state transition probabilities p_i .

$$\begin{aligned} g_1(p) &= c_1 \\ &\vdots \\ g_m(p) &= c_m \\ h_1(p) &\leq c_{m+1} \\ &\vdots \\ h_n(p) &\leq c_{m+n} \end{aligned} \quad (13)$$

²To obtain the maximum of $f(p)$, the minimum of $-f(p)$ has to be calculated.

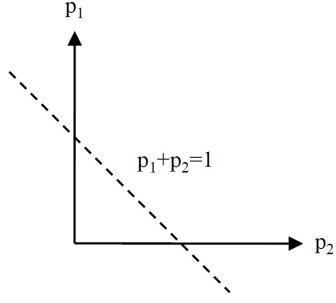


Figure 5: Equality constraint, limiting the solution space by an affine subspace.

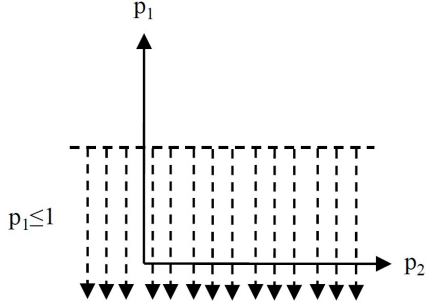


Figure 6: Inequality constraint, limiting the solution space by a halfspace.

A polyhedron can be defined as a system of linear equations. To obtain such a polyhedron, the inequality equations $h_l(p)$ must be transformed to equality constraints by inserting additional variables s_l for each inequality:

$$h_l(p) \leq c_l \rightarrow h_l(p) + s_l = c_l, s_l \geq 0 \quad \forall l \in [(m+1) \dots (m+n)] \quad (14)$$

This leads to the following system of linear equations:

$$\begin{aligned} g_1(p) &= c_1 \\ &\vdots \\ g_m(p) &= c_m \\ h_1(p) + s_1 &= c_{m+1} \\ &\vdots \\ h_n(p) + s_n &= c_{m+n} \end{aligned} \quad (15)$$

with the solution vector $x = (p_1, \dots, p_k, s_1, \dots, s_n)^T$. This can also be written in matrix form for a $(m+n) \times (k+n)$ -Matrix $A = (a_{i,j}) \in \mathbb{R}^{(m+n) \times (k+n)}$:

$$A \cdot x = c, \text{ where } c = (c_1, \dots, c_{m+n})^T \quad (16)$$

It is obvious that

$$k + n \geq m + n \quad (17)$$

holds since the dimension of the solution space (k) is always greater than or equal the number of linear constraints (m). For each linear constraint, the solution space is limited by

a further affine subspace and therefore loses (at least) one degree of freedom. In the special case where $k + n = m + n$, the vector x is implicitly given and no further inequality constraints can affect the result. Let b_r^s be the r -th basis vector in column s of A defined as

$$b_r^s = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_{m+n} \end{pmatrix}, \quad \text{where} \quad \beta_r = 1, \beta_i = 0 \quad \forall i \neq r. \quad (18)$$

Assume that

$$(\alpha_1, \dots, \alpha_{m+n}), \alpha_i \in \mathbb{Z}^+ \quad (19)$$

is a possible arrangement of basis vectors within A and α_k is corresponding to $b_k^{\alpha_k}$. All possible combinations of basis vectors are given by the set

$$\mathcal{B} := \left\{ (t_1, \dots, t_{m+n}) \mid t_i \in \{\alpha_1, \dots, \alpha_{k+n}\}, t_i \neq t_j \right. \\ \left. \forall i, j = 1 \dots (m+n) \right\}. \quad (20)$$

Each element $z_i \in \mathcal{B}$ represents a combination of basis vectors, which leads to a valid basis solution of A . To obtain a solvable order of these basis vectors, one can choose a permutation of this combination for which A is solvable. The set of possible permutations for a given combination of basis vectors is given by the set

$$\mathcal{P}(z_i) := \left\{ (t_1, \dots, t_{m+n}) \mid t_i \in z_i \setminus \{t_1, \dots, t_{i-1}\} \right\}. \quad (21)$$

Each basis solution x_i represents an intersection of affine subspaces and half spaces, while basis solutions with $x_i > 0$ represent the corners of the convex polyhedron, marked with black dots in Fig. 7.

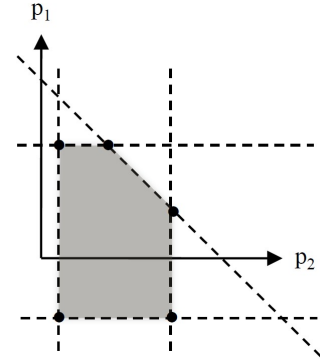


Figure 7: Valid solution space with respect to constraint definitions.

Let $\{s_i\} \subset \mathcal{B}$ be the set of w unique permutations of z_i which lead to solvable basis solutions x_i with $x_i > 0$ for $i = 1 \dots w$. The polyhedron itself represents the valid solution space given by the defined constraints. Every point within this solution space leads to a valid probability distribution (see Fig. 7). Of course, one of these valid solutions must be picked now and there might be a multitude of possible approaches for doing so.

Intuitively, a probability distribution that matches the semantics of the defined constraints must be found somewhere in the center of the polyhedron.

A suitable approach for choosing a distribution is to calculate the polyhedron's *center of mass* $\mathbf{R}$ as follows (see also Fig. 8).

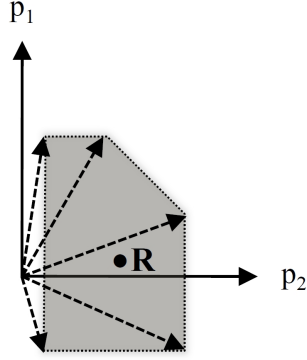


Figure 8: Center of mass of a Polyhedron.

Let x_i^p be the vector of k upper elements of x_i , with $x_i^p(p_1, \dots, p_k)$, which holds the searched probability distribution. The center of mass $\mathbf{R} \in \mathbb{R}$ is given by

$$\mathbf{R} = \frac{1}{w} \cdot \sum_{i=1}^w x_i^p \quad (22)$$

In Table 1 and Table 2, we compare the solutions for both polyhedron and maximum state entropy approaches for different constraint definitions.

Table 1: Maximum Entropy vs. Polyhedron approach for the constraint $p_1 \geq p_2 \geq p_3 \geq p_4$.

	p_1	p_2	p_3	p_4
Max. Entropy	0.25	0.25	0.25	0.25
Polyhedron	0.52	0.27	0.15	0.06

Table 2: Maximum Entropy vs. Polyhedron approach for a constraint $p_1 \geq p_2 + p_3$, $p_3 \leq p_4$.

	p_1	p_2	p_3	p_4
Max. Entropy	0.36	0.18	0.18	0.27
Polyhedron	0.45	0.13	0.09	0.33

In our opinion, the transition probabilities obtained from the polyhedron approach (22) are more closer to the intended constraint semantics than the probabilities that are derived by maximizing the state entropy function (6). Instead of delivering values that are always as uniform as possible, in other words as random as possible, our polyhedron approach is focussing on the main area of all constraints inequalities. As a result, we will not always get marginal probabilities for the Markov chain usage model.

It should also be remarked that in the limiting case of using only equalities both techniques produce the same solution set for the transition probabilities.

4.3 Constraint Modeling

For reasons of compactness and to avoid inconsistencies, constraints can be placed directly within the model. The

combination of both, constraints and models, result in a MCUM usage profile.

For the proprietary Siemens tool TDE, developed by Siemens Corporate Research, models are represented by extended finite state machines and constraints can easily be placed as simple text fields, as depicted in Fig. 9.

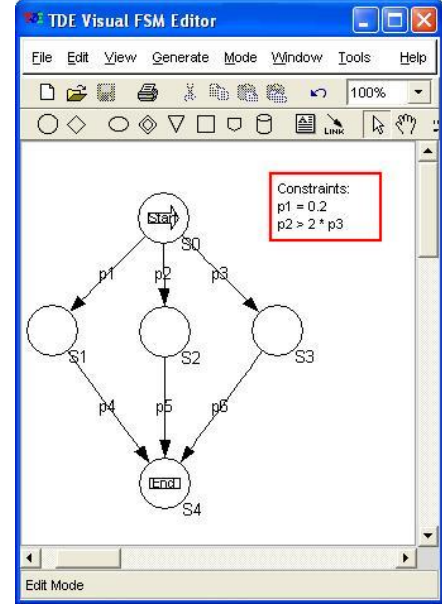


Figure 9: Modeling of constraints in TDE.

5. USAGE PROFILE COMPARISON

"How much does the expected usage of two customers differ?"

Usage profiles may be compared³ with respect to their difference in the average usage of the SUT. Resulting values of the wanted metric indicate the magnitude of the deviation between two usage profiles. Thus, high values prove that two customers use the system very differently.

The KLIC (Kullback-Leibler Information Criteria [Pro00]) is an information theoretic metric for revealing the difference between two probability distributions P and Q and is expressed by the Kullback-Leibler divergence:

$$KL(P, Q) = \sum_i p(i) \cdot \log \left(\frac{p(i)}{q(i)} \right) \quad (23)$$

In order to compare Markov chain usage profiles, this metric is extended in the following to consider steady state information, i.e. the rate $\pi(s_i)$ a user will chose state s_i of P during the average system usage:

$$KL(P, Q) = \sum_i \pi(s_i) \sum_j p_{i,j} \cdot \log \left(\frac{p_{i,j}}{q_{i,j}} \right) \quad (24)$$

Evidently, $KL(P, Q)$ is not symmetric, i.e. in general,

$$KL(P, Q) \neq KL(Q, P) \quad (25)$$

³Usage profiles have to be based on the same MCUM structure to be comparable.

To calculate the KLIC it is assumed that P represents a precisely calculated probability distribution, whereas Q is an approximation of P . Since usage profiles have to be treated equally, some kind of average KL divergence is necessary.

The Jensen-Shannon divergence (JSD) is a symmetric metric for calculating the similarity between two equally treated probability distributions and is used to determine the deviation between two MCUM usage profiles P_1 and P_2 .

$$JSD(P_1, P_2) = \frac{1}{2}KL(P_1, M) + \frac{1}{2}KL(P_2, M) \quad (26)$$

with

$$M = \frac{1}{2}(P_1 + P_2) \quad (27)$$

Table 3 shows the variation between three usage profiles that are automatically generated from a MCUM in order to test a medical workflow [Hol08].

Table 3: Deviations between usage profiles.

	General Profile	USA Profile	Australian Profile
General Profile	0	0.1586	0.1242
USA Profile	-	0	0.1612
Australian Profile	-	-	0

In the general profile, usage probabilities within medical workflows are averaged over hospitals worldwide, whereas the USA and the Australian profiles characterize usage probabilities for country-specific hospitals. A zero value indicates no profile difference. It can be seen that an Australian hospital type is more closer to an average hospital than the USA one.

Another possible application of divergence measurements between MCUM profiles is the classification of customers. A customer's profile could be compared and assigned to concrete user classes, based on the divergence of the usage profiles. For instance, such a class could be "3D User" that is identifying customers who use 3D features of the software extensively. After a customer is assigned to a class, comparative data emerges for a variety of parameters. A simple example is the customer's hardware standard. Customers in the class "3D User" are likely to have similar standards for their hardware equipment. Furthermore, customer trainings could be tailored to these classes to give a certain group of customers specific trainings for the SUT.

6. TEST PLANNING

We finally discuss a technique based on the KL divergence, which allows to predefine the number of test cases that have to be executed. This issue is of main interest if the time for test case execution is limited at the end of a software development project.

In this situation, the KL divergence can be used to identify the quantity of coverage needed between sampled test cases and the expected usage of the system given by a MCUM profile. Thus, a low value (e.g. below the threshold value 1 in Figures 10 and 11) states that testing matches the expected customer usage behavior very closely and can be stopped.

Let s_i be an arbitrary state with n outgoing arcs $a_{i,j}$. Let further $k_{i,j}$ denote the number of times the transition $a_{i,j}$ is covered by a set of sampled test cases. The resulting coverage rate $t_{i,j}$ for the test profile is calculated as

$$t_{i,j} = \begin{cases} \frac{k_{i,j}}{v_i}, & \text{for } v_i \neq 0 \\ 0, & \text{for } v_i = 0 \end{cases} \quad \forall j = 1..n, v_i = \sum_{j=1}^n k_{i,j} \quad (28)$$

Let T denote a test profile resulting from a test case sample and U an arbitrary usage profile. The KL divergence is defined as

$$K(U, T) = \sum_i \pi_i \sum_j u_{i,j} \cdot \log \left(\frac{u_{i,j}}{t_{i,j}} \right) \quad (29)$$

Evidently, the KL divergence is defined only if all arcs of the usage model are tested at least once. To circumvent this situation, [Say99] proposes an approximated $\overline{KL}$ divergence. For arcs, which are not covered by test cases, $t_{i,j}$ is substituted by a small value $\epsilon > 0$ that is close enough to zero depending on the wanted accuracy in order to avoid a division by zero. The resulting approximated $\overline{KL}$ divergence is given by

$$\overline{K}(U, T) = \sum_i \pi_i \sum_j u_{i,j} \cdot \log \left(\frac{u_{i,j}}{\epsilon - \epsilon \cdot (\text{sgn}(t_{i,j})) + t_{i,j}} \right)$$

with

$$\text{sgn}(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ -1, & \text{if } x < 0 \end{cases} \quad (30)$$

Figures 10 and 11 explain the behavior of the $\overline{KL}$ divergence during test case sampling with respect to different usage profiles. The X-axis is showing the number of test cases already used, whereas the Y-axis represents the corresponding $\overline{K}(U, T)$ values. Obviously, the $\overline{KL}$ divergence tends to reach lower values for samples from usage profile from which the test cases are sampled.

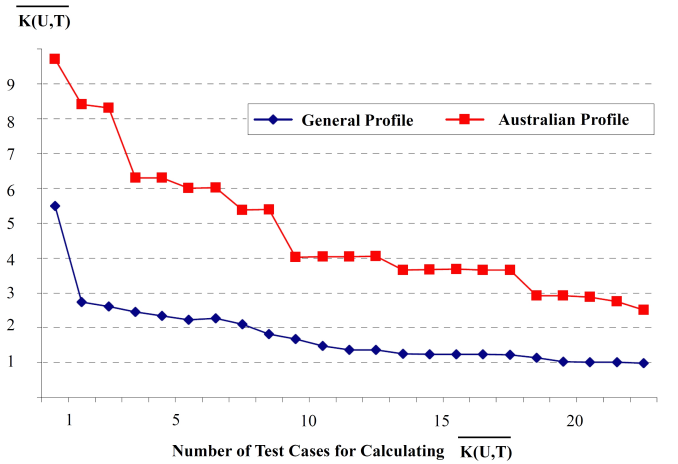


Figure 10: Test cases sampled from General Profile.

7. CONCLUSIONS

We have shown that statistical testing based on MCUMs is a promising extension to existing deterministic testing approaches in general, and in the medical domain in particular.

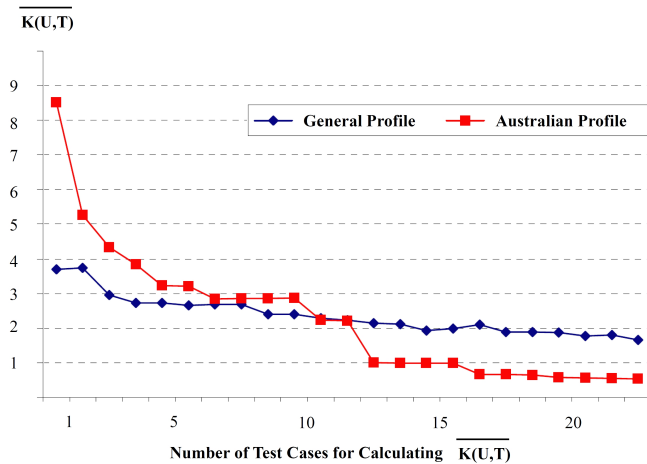


Figure 11: Test cases sampled from Australian Profile.

In order to define customer usage profiles, appropriate transition probabilities have to be obtained. The constraint-based approach turned out to be most convenient for this purpose. A polyhedron method for calculating suitable probability distributions has been presented and compared to the maximum entropy approach.

From the usage models instructive metrics, we have not explicitly mentioned, can be analytically derived to improve test planning or to support management decisions. Examples are the mean frequency a customer will chose certain usage states or the mean lenght of the test cases for estimating the test exection time. Specific scenarios for comparing different customer classes by means of dedicated MCUM usage profiles have also been identified and discussed.

As for test planning, a method based on the KL divergence to integrate a practicable test stop criteria into existing test case selection procedures has been proposed.

In summary it can be said, model-based test case generation using MCUMs and dedicated profile classes enable promising techniques for testing complex systems, for example in the medical or the automotive domain. In the latter case, we are currently extending the test framework of a leading German automobile manufacturer in order to integrate statistical test case generation techniques [SDGK09].

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